

## Myth, Reality and the Future in IP Telephony

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**PBXL** is a venture company focused on delivering integrated IP Telephony solutions to small and mid-sized companies in Japan and beyond, focused on delivering the same quality of service that the big companies get, without having to spend tens of thousands of dollars on an acceptable phone system.

This essay outlines the story of IP telephony, the **myths** of the past, **today's reality**, predictions for the **future**, plus various other entrepreneurial opportunities that remain for the savvy businessperson.

To make sure we're speaking a common language, we will start with a few definitions and clarifications. This essay uses the following terms: **Voice over IP**, which is a technical way in which voice sounds get from one end to another. The IP network could be a private carrier, a LAN within an office, or the Internet. **Voice over the Internet** is a variation of VOIP, where the voice travels across the public Internet. **IP telephony** is the term used today to describe the general case of convergence enabled by IP, and combining computer telephony integration with other types of voice offerings.

**Myth 1: IP Telephony is cheap or free.**

- IP Telephony may ride for free, utilizing existing network capacity but external calls require connecting to the conventional telephone network.
- Many companies focus on the wrong series of charges—voice per minute cost is small compared to PBX equipment and base line charges. In fact, in several of the companies that we have worked with, they so overwhelm per-minute costs as to make usage costs nearly irrelevant. In one case, one of Weisser Consulting's [CommDown](#) clients had a 17 year ROI on a new VOIP installation, with the upgrades required to their circuits, WAN and new equipment, based on the savings they would receive from moving to VOIP.

**Myth 2: IP Telephony is unreliable.**

- There is nothing inherently unreliable about an IP network. However, most of these networks are designed differently than voice networks. Voice Over the Internet, is probably less reliable than a dedicated network.
- We have also found that combining voice and packet networks within an office can lead to some degradation in sound quality, though VLAN management can address the bulk of these issues.

**Myth 3: IP telephony spells the death of distance and of time-based billing.**

- While we see consumer services in the US that are fixed fee for all domestic (and some international) calls, Japan only has distance-insensitive billing, at least for off-net calls. That may change over time, but it has taken years longer than expected, at least by those of us who thought that we would see the end of such billing by the end

of the last century. Since wholesale billing continues to be on the per-minute rather than the per-Meg basis, this will take a long time to evolve.

**Myth 4: (implied) Consumers and businesses have the same needs with telephony.**

- Telephony for groups is radically different than for individuals, as evidenced by things like hunt groups, call forwarding, call routing within an organization, line trunking and least cost routing.

[During the actual talk, there was an extended discussion when I asked for additional IP telephony myths. Consensus was that the other two myths that needed to be addressed were “IP Telephony is insecure” and “IP Telephony is necessarily of poor quality”. For the former, since it is actually possible to encrypt IP telephony end to end, a compelling case can be made for IP Telephony *potentially* being more secure, at least in certain ways, than conventional telephony. In fact, in our experience, conventional PBXes often have remote maintenance lines open which are only lightly monitored. Quality is a different series of issues altogether, due to the wide range of services available to someone implementing IP Telephony. For example, while a conventional voice call requires 64 kbps, we see certain IP solutions which can fit between four and eight calls within that same amount of bandwidth. Such compression reduces the voice quality, but allows for more simultaneous calls on the same line. In the case of a dedicated IP line with superior switching infrastructure, we see voice quality that is sometimes **too good**—with no background hiss, our customers sometimes wonder if the connection is still up. For a more complete discussion, see the PBXL web-site article “Comparisons of Mixed IP/VOIP implementations”

## Reality

The consumer IP Telephony market is growing more mature in terms of reach, and is heading towards richness of service. Observing products in the market generally, successful consumer services have the following characteristics:

- **They are easy to use.**
- **They integrate well with something that a consumer is already used to. An example here in Japan would be the i-Mode service when NTT introduced it. It was cheap, easy to use, and had compelling content despite being delivered over the cell phone.**
- **They are cheap but are clearly best-effort services.**
- **They require minimal hardware expenditure, but instead operate on a service model, so customer life-time value justifies potentially high acquisition costs.**
- **Their designs are such that most users utilize 70% or more of available features.**

Offerings such as Skype or YahooBB immediately come to mind when exploring IP Telephony in the consumer market. Skype is an interesting case—utilizing **peer to peer connectivity**, it is an exciting example of modern **voice over the Internet** capability. Skype also uses encrypted connections, recognizing the security concerns inherent in a peer to peer, voice over the Internet. They have also done interesting things with integration into the conventional, switched voice telephony network, with their Skypeout and Skypein products. What Skype has not done is address the needs of the typical business user of phones, who are

**(a) not comfortable sharing their corporate network with other traffic**

**and (b) prefer a phone that doesn't require a PC.** Additionally, with the current state of the Skype offering, there is no way to ensure connectivity to other networks.

Having built a **softphone** service option into **PBXL**, however, we can see that the cost of supporting **PC-based telephony** as anything other than a “**value-added option**” is very high. Softphone, in our experience is not acceptable for many users because businesses want **telephones** and not more PC support issues. YahooBB avoids this issue altogether, of course, by not providing any such options. That, however, brings us to our second, unique to Japan case.

A couple of years ago, Yahoo crashed the market with an integrated voice and data offering with DSL. This was brilliant, and they were the drivers of broadband adoption here—an analyst's report in 2001 predicted that in 2003 or 2004 DSL would start to decline as more people moved to fiber-optic lines. As it turns out, that move has only begun recently, and the number of DSL subscribers still outnumber the number of fiber users significantly, and probably will for at least another two years at the current rates of growth, since there is no incremental advantage to utilizing fiber over DSL to the consumer.

YahooBB, though, has clearly had more success in the **consumer market** than in the business market, even with the Softbank acquisition of Japan Telecom and Cable and Wireless IDC. Most of the group's products however, have been focused more on *lines* than on *services*, which is typical of the carrier mind-set, and a key weakness in the telephony offerings here in Japan.

Carriers generally think in lines and minutes, which is partly why they have had trouble adjusting to things like web services. Even now **the**, traditional carriers still separate their voice sales staff from their data sales staff—which becomes less relevant by the day. **Old-line carriers**, though, are very concerned with IP telephony, as circuit-switched voice has been lucrative traditionally but is largely commodified. After all, **by merely changing a prefix on your PBX** it is possible to automatically re-route to a different network. This is hardly the “**sticky relationship**” that is desired in any business.

Looking at the market in Japan, I believe that there **few particularly good options** for mid-sized businesses in the hardware and telephony market, and particularly not in the general marketplace. I believe this for three reasons:

1. **The PBX market in Japan is dominated by a few players** who control most of the distribution and sales. The business models of these companies have been designed around large, one-time sales and then minimal revenue “maintenance contracts” post-sale.
2. Most of the purchasers in Japan have traditionally been **unsophisticated**, often without the involvement of the IT department, and rarely with a focus on functionality and expectations. IT departments also generally do **not** like maintaining the PBX either, as it traditionally has not been a piece of equipment they know how to manage, or it requires the hiring of a dedicated or

semi-dedicated staff member to manage occasional move/add/changes.

3. The functionality of existing consumer services do not even come **approach** the options available in a high end PBX, though they are similar to what you see in the “key systems” on the market.

### **Historical PBXes and Business phone systems**

PBXes, have the following characteristics:

- They are easy for the **end user**.
- They require **significant additional integration effort**.
- They are very **expensive** to purchase and/or to upgrade.
- They **require space** and need to be in **cool, temperature-regulated** environment.
- **Service integration is rarely given significant thought**.
- The feature design is rich enough that **most users utilize only a small percentage** of their possible options.

Traditional PBX vendors refuse to seriously consider Skype or YahooBB because they have dominance in the feature market. **Modern PBXes are rich in features and functionality, but the cost to a company is extraordinary.** This is in part due to legacy style design (based on numbers like 23, 155 and 622 instead of powers of 10 or two) and in part because of lack of competition in the market. A typical, feature rich and functional PBX in Japan might run a 30 person office **between 4 and 8 million yen**, while at the same time offering a “maintenance contract” that

charges you every time you make a maintenance call. In the pre-IP days, companies needed a PBX for each office, leading to a disparate collection of “very cheap” PBXes in the satellite offices, with a “real” PBX in the main office. Now, due to some of the distributed intelligence available via IP PBXes, you no longer have to buy separate phone switches for each office, but the hardware and software remains expensive.

Additionally, unlike the computer and server world where we see better and faster equipment all the time, **each little feature on a PBX costs money**, be it more hard disk space (priced in 1990s hard disk dollars), IP functionality and “hold music”. **Cheaper PBXes (i.e. the domestic phone manufacturers) seem to provide even less for the money**—many of the users we have talked with settled for a domestic manufacturer and despised it ever since. Additionally, at the point you purchase a PBX, you literally cannot do anything with it. Instead, you (or sometimes the company that sold you the PBX) will have to order a bunch of lines from your favorite local phone company, which, based on my current experience opening a new office, still acts like a monopoly when it comes to customer service. Finally, there was a time when 10 hours of voice mail storage would cost you 40,000 US Dollars, despite the fact that you could get a hard disk at the time for about 100 USD. That situation has improved marginally, but **awaits real competition**.

All that being said, there are some **big changes coming to the business voice/PBX market**, of which open source platforms like Asterisk are part of the solution. It is certain that **hosted PBXes** are going to seriously disrupt



how both traditional carriers and hardware companies do business. In the mid- 1990s, I sold and delivered lines to companies who were putting up Sun Microsystems workstations in their offices for web servers, increasing the IT overhead and incidentally creating problematic security issues. Yet, by the end of the 1999, most customers **had migrated to a secure data center for hosting purposes**. In the same way, we think that **hosted PBXes** are going to make a similar move over the next five years or so. We're not alone—**Yankee Group** predicts this to be a **three billion USD market by 2010**. Anecdotally, it's clear to me that some of the IT companies do not understand the difference—in the course of introducing PBXL to a number of potential partners, I received a response from one of them along the lines of “we already sell Cisco phones.” On the carrier side, the most common response has been “we offer IP Centrex and haven't had any luck.” Centrex offerings by telecommunications companies date back to the **1950s**, where a customer organization essentially purchases PBX-style services via the carrier's switch. The problem with many of these offerings, however, is that they require very large organizations for there to be any cost benefit, as set-up and maintenance **on a traditional switch is very, very expensive**, and typical sales personnel for the phone companies were not adept at selling these services. So, it is important to differentiate Centrex (and IP Centrex) solutions as a **carrier-PBX offering**, rather than a hosted PBX, which are focused more on small and mid-sized businesses, and provide service bundles.

Hosted PBXes generally **combine all of the hardware, software and voice offerings into a consolidated service**, and may offer additional services such as voice mail to e-mail, a.k.a unified messaging (either as an

attachment or a link), and other types of conventional telephony services. In our particular case, with PBXL, we introduced a combined voice and data offering, with distance-insensitive voice service, a built-in direct dial number and all of the services customers expect in a phone system, such as 3-way calling, voice mail, call forwarding. Further applications are what will drive the market, however, which means we should start with some of the largest purchasers of PBXes and PBX services, **call centers**.

Call center applications have already made a great deal of real progress in **telephony related integration and CRM**. Call centers use voice and applications better than any other other business, with better integration into back-office applications to minimize the amount of time that customers have to spend on the phone with a customer support person because it is a key determinant of how much money they can make. **This isn't to say that these applications are often well implemented**—in too many cases, we find interactive voice response systems which aren't very interactive and don't respond the way they are expected. Beyond this, however, common applications include using number display/caller-ID to identify the incoming caller, pop-up a screen with their personal information, and enable better service.

**That's the old technology, though.** Some of the newer innovations over the past couple of years have enabled call centers to move from a “cheap location in the countryside” or to “different countries with lower cost structures” **to a combination of “a section of the office” and “stay-at-home workers”** through IP Telephony, integrated with real-time **chat and fuzzy logic chatbots** that are the IVRs of cyberspace. Large multinationals have already incorporated many of these technologies into their outsourcing

plans, and we see smaller companies attempting to make the move as well, (to the extent that we have already answered an RFP for a **Japan-Philippines outsourcing connection**.) And while PBXL is only at the “flexible location” component of the call center, we’re working on the CRM integration requirements. **Finally, with more PC-based interaction with the customer**, it is easier for video conferencing to succeed, as there are many more cameras connected to PCs than videophones. Call centers have led the way in showing us what kinds of things might be possible in the world of business telephony, but we are just now starting to see the **actual deployment of directory applications** on a common basis for business users.

Business roll-outs generally have a different focus than call centers, with an interest in **contact management**, **internal and external directories**, and other types of phone service. Additionally, **unified messaging**, a feature which does not matter much in the world of contact centers, is finally moving to the forefront of the feature list—about five years after some of the entrepreneurs in the market began offering it. Beyond that, however, in the same way that call centers are becoming contact centers with **computer to computer** enabled messaging, integrating contacts from Outlook or from Notes into a phone system can add significant value and is moving up on the “want-to-have” list of features, as it is not yet on the “required” list. Unfortunately, the amount of **hardware and software** that companies need to buy to implement this functionality can run in the several tens of thousands of dollars, making it impractical for

a small business to implement. At the same time, however, Cisco worked with us to architect something that would still be effective for small businesses, spreading the cost of an integrated PBX across different companies and giving us economies of scale that individual company purchases don't have. We are also working with them to better address desires and needs of companies **within wireless space**—in my experience doing cost reduction work for companies here in Japan, the cell phone bills of a heavily-mobile sales force will easily outweigh the costs of fixed line service, so let us briefly examine the topic.

There are actually two parts to the wireless equation, **the local office wireless** connectivity option and the “**outside the office**” wireless option, using mobile phone carriers. For the local office connectivity option, we have had relatively expensive technology in place for several years, starting with **PHS** and moving now to the dual-mode **802.11** phones. These phones allow mobile phones to use existing in-office wireless networks and connect to the local PBX, lowering the cost of usage for calls made within the same “local” PBX. The pricing of the PBX add-on equipment, however, requires a fairly detailed analysis of network usage prior to purchasing the option, similar to the 18 year ROI mentioned above. For outside the office integration, while it is **technically possible to provide some sort of integration into a PBX**, there are currently no mobile offerings from Japanese carriers that allow for this sort of integration. Any observer of the US and European markets, however, would note the success of the Blackberry, which has provided integration a mobile device back to a server and also notice that Japan doesn't seem to have a competing service, though Visto is working on obtaining similar access from our cell phones.

[Question and Answer at this point was much less lengthy, making it clear to the author that cell/data convergence is still very, very early]

Looking at the progression of telephony to IP Telephony, and the integration of services of the future, I believe that the following will be key drivers of progress.

1. **Integration into a PC, server and Internet connected environment**, including content distribution to the phone.
2. **Integration into applications on demand**, such as Salesforce.com and Netsuite.
3. **Integration** into public **wi-fi** and possibly other types of wireless networks.
4. Improved **collaboration**, with the ability to record all calls and archive at a negligible cost.
5. **XML** compatible phones are will be a supplement to PCs on the desktop, but may eliminate the PC altogether for call center applications.
6. **Security**, although it has not been extensively addressed so far, is a growth area both for IP Telephony and applications and will remain relevant. In PBXL, we have implemented all of our IP Telephony connections via a VPN, which addresses most of the issues on the

network side, but of course it is important to realize that the security issue isn't going to disappear.

From an entrepreneurial approach, I think that anyone who focuses on making useful applications, particularly cross-platform (e.g. Salesforce.com and Skype integration) has the opportunity to be very successful.

Some things that get lots of coverage but I am more skeptical about:

Presence. Individual availability in the near-term won't be terribly successful, at least not for typical business applications, as most people like the flexibility of not responding to every request. Call Center applications will be different, of course.

Dedicated VOIP providers who don't provide meaningful service integration. It is too easy to switch providers with a minimum of effort, thus the provider business becomes a cost-driven, commodified experience.

In the course of this essay we have covered some of the myths related to IP Telephony, the reality of where the market is today and some of the possible avenues that are open in the future, particularly focusing on integrated offerings. There are certainly other opportunities which have not been discussed here, other myths to address and a good deal of reality that we haven't touched on, but I hope that the reader has now been provided some food for thought, and would welcome comments.